

CSMPRO00000004 Pittsburgh Bridge Retrofit Proposal

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 **Category:** Infrastructure / Pittsburgh

Revision Drivers (all V2.0 documents)

1. **2025-2026 experimental and regulatory data** supersedes V1.0 specifications
 2. **Solar Cycle 25 SSN ~137 peak** (observed) — elevated urgency for all Carrington-focused work
 3. **Standards updates** — AASHTO 2024, IMO 2025, IEEE P2945 draft, DNV GL EMP-2, ClassNK FRP
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V1.0 vs V2.0 Parameter Changes

Parameter	V1.0	V2.0	Consequence if Not Updated
Fern Hollow bridge	Not mentioned in V1.0	V2.0 integrates Jan 2022 collapse forensic data	Major local case study missing from V1.0 reduced proposal credibility
GIC amplification data	Appalachian Plateau qualitative	2.8x confirmed by geological survey	V1.0 lacked quantitative regional amplification factor
IIJA funding	Mentioned generally	BRR \$26.5B; PA allocation \$870M; \$13.6M IIJA match on CSM program	V1.0 missed specific federal funding amount
Three Rivers analysis	General GIC threat	V2.0 Fort Pitt Bridge: 43,000 A; P=555 kW; 14-min failure	Specific failure time missing from V1.0 — reduces stakeholder urgency
NFL Draft urgency	Not in V1.0	2026 NFL Draft: 250,000 visitors; bridge criticality demonstrated	V1.0 missed time-sensitive political/commercial opportunity

Unchanged Elements

All core design philosophy, threat physics (GIC induction equations), material selection rationale, and fundamental architecture are carried forward unchanged from V1.0. Only the parameters listed above were revised.

End Versioning Delta | CSMPRO00000004 Pittsburgh Bridge Retrofit Proposal | June 2026